Store Staffing Analyzer

Retail Staffing Analysis

Data-driven staffing optimisation for a retail store

EXECUTIVE HEADLINE

Hire Immediately (8 from month 1) - moving from 7 to 8 heads adds Rs.8.08 L of gross margin over the next 6 months versus leaving the vacancy unfilled.

Report detail	Value
Dataset	RevI-Test
File	RevI-Test.csv
Rows / columns	81 / 18
Months of history	81
Salespersons	16
Total sales	Rs. 8,375.79 L
Generated	2026-10-22 18:26:30
Run created at	2026-10-22T12:56:18.714390+00:00
Run id	c4ae7ae158c2
Schema version	2.0
Assumptions	Salary 3.00 L, commission 5.0%

All monetary values are in Lakhs INR. This report was generated automatically by the Store Staffing Analyzer analytics engine; every figure is read directly from the analysis payload.

Table of Contents

1. Executive Summary
2. Financial Highlights
3. Forecast Outlook
4. Staffing Recommendation
5. Scenario Comparison
6. Productivity & Diminishing Returns
7. Risk & Sensitivity
8. Payback & ROI
9. Business Rule
10. Appendix

Each section presents the KPI tables, the analytical detail and the supporting charts for the corresponding part of the staffing decision.

1. Executive Summary

Headline

Hire Immediately (8 from month 1) - moving from 7 to 8 heads adds Rs.8.08 L of gross margin over the next 6 months versus leaving the vacancy unfilled.

CONFIDENCE: MEDIUM

Recommendation at a glance

Metric	Value
Recommended action	Hire Immediately (8 from month 1)
Recommended scenario	B
Confidence level	MEDIUM
Runner-up option	Scenario D: Conditional Hire (trigger >= 114.09 L sales)
Current employees	7
Capacity-optimum headcount	9

Expected 6-month financial impact

Metric	Value
Total sales (6 months)	Rs. 702.52 L
Total gross margin (6 months)	Rs. 172.13 L
Total staff cost (6 months)	Rs. 179.13 L
Average monthly gross margin	Rs. 28.69 L
Margin advantage over runner-up	Rs. 0.00 L
Margin advantage vs do-nothing	Rs. 8.08 L

Confidence drivers

Metric	Value
Sales consistency (CV)	0.2524
Historical months	81
Headcount observations	4
Margin gap to runner-up	0

Confidence is driven by sales consistency (coefficient of variation), the depth of history and how clearly the winning scenario separates from the runner-up.

2. Financial Highlights

Key performance indicators

Metric	Value
Current employees	7
Capacity-optimum headcount	9
Average monthly sales	Rs. 103.40 L
Average monthly gross margin	Rs. 25.42 L
Total gross margin	Rs. 2,059.11 L
Best monthly gross margin	Rs. 44.35 L
Worst monthly gross margin	Rs. 1.39 L
Sales volatility (std. dev.)	Rs. 27.58 L
Break-even sales per employee	Rs. 6.67 L
Average sales per employee	Rs. 14.69 L
Productivity variability (CV)	0.252
Forecast average monthly sales	Rs. 107.43 L
Forecast average monthly margin	Rs. 28.69 L
Best forecast model	Ensemble (Top 3)
Validation RMSE	Rs. 8.02 L
Validation MAPE	5.9%
Validation MAE	Rs. 5.84 L
Validation sMAPE	5.7%
Validation R2	0.846
Diminishing returns detected	Yes
Total staff cost ({m} months)	Rs. 179.13 L
Months to payback	1.00
Return on investment	44.9%
Anomalous months detected	1
Data completeness	1.0%
Confidence level	MEDIUM
Seasonality strength	0.9%
Detected seasonal cycle (months)	6
Historical leavers	9
Marginal sales per employee	Rs. 11.25 L
Marginal margin contribution	Rs. 2.06 L
Marginal model R2	0.890

Margin formula

Gross margin = (1 - COGS - commission) x Sales - salary x employees, i.e. 0.45 x Sales - 3.00 x employees, with salary Rs. 3.00 L per employee per month and commission 5.0% of sales.

Monthly gross margin - most recent 24 of 81 months

Month	Sales	Active employees	Sales / employee	Gross margin
58	119.52	7	17.07	32.78
59	121.59	7	17.37	33.72
60	123.46	7	17.64	34.56

Month	Sales	Active employees	Sales / employee	Gross margin
61	116.24	7	16.61	31.31
62	94.57	8	11.82	18.56
63	61.21	7	8.74	6.54
64	122.80	7	17.54	34.26
65	125.00	7	17.86	35.25
66	99.40	8	12.42	20.73
67	108.85	7	15.55	27.98
68	111.19	7	15.88	29.04
69	67.93	7	9.70	9.57
70	115.22	7	16.46	30.85
71	117.02	7	16.72	31.66
72	118.72	7	16.96	32.42
73	118.89	7	16.98	32.50
74	96.05	7	13.72	22.22
75	61.43	7	8.78	6.64
76	130.94	8	16.37	34.92
77	112.56	7	16.08	29.65
78	101.40	7	14.49	24.63
79	116.68	7	16.67	31.50
80	118.45	7	16.92	32.30
81	69.61	7	9.94	10.32

Note

Showing the 24 most recent months of 81 available months of history. All values are in Lakhs INR.

Cost of an open position

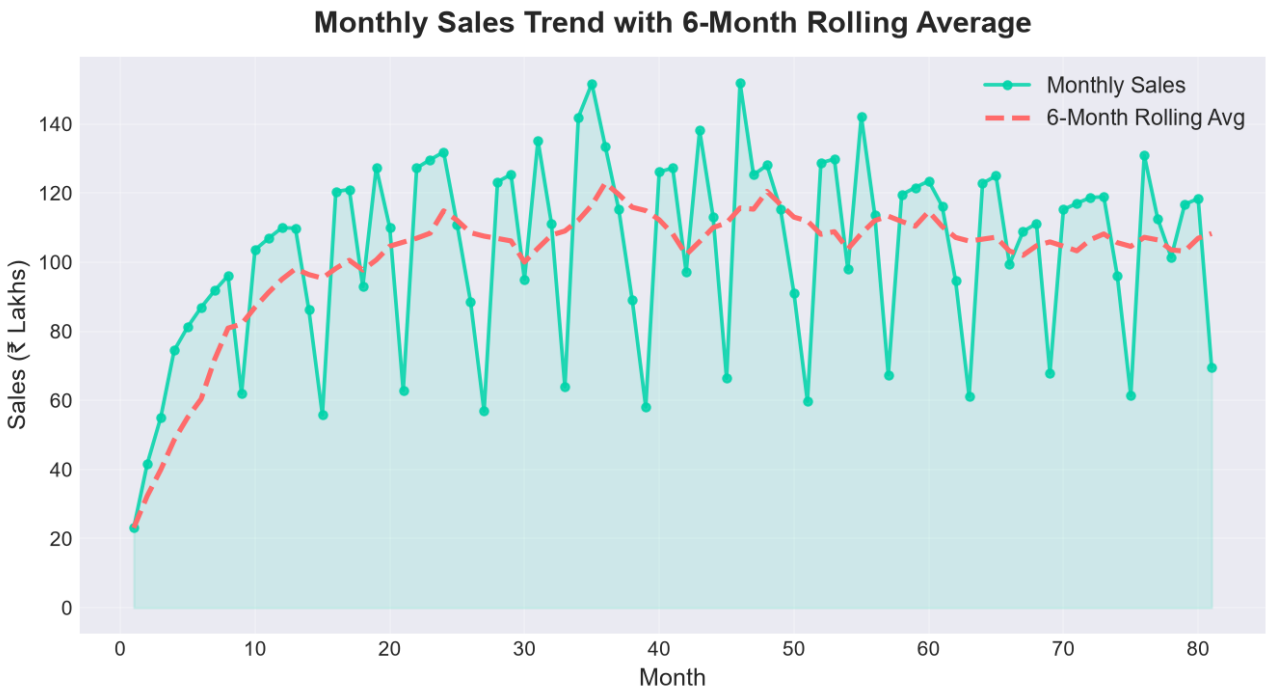
Leaving 1 position(s) open is estimated to cost Rs.0.85 L of gross margin per month (Rs.10.24 L annually), against a salary saving of Rs.3.00 L per month. Net monthly impact of hiring: Rs.-2.15 L.

Seasonal demand profile

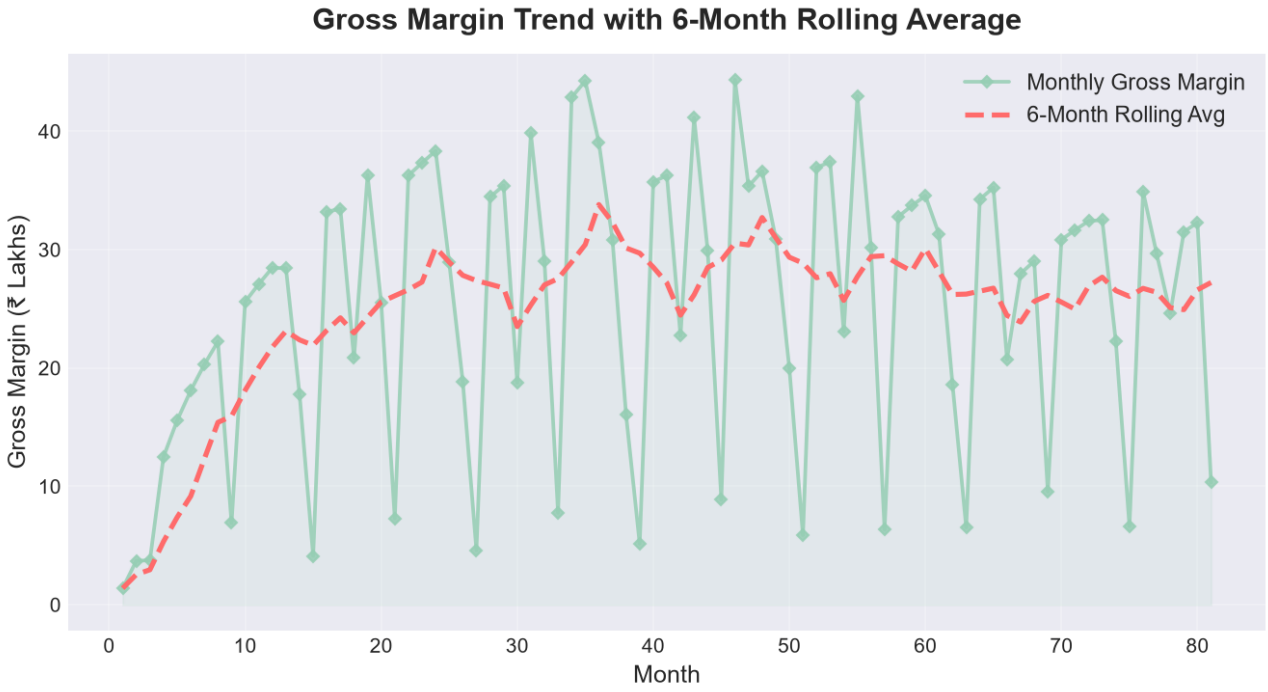
Month	Name	Index	Strength	Average sales
1	Jan	1.066	0.000	101.36
2	Feb	0.844	0.000	83.86
3	Mar	0.545	0.502	58.32
4	Apr	1.146	0.000	118.10
5	May	1.157	0.000	117.47
6	Jun	0.887	0.000	95.83
7	Jul	1.183	0.658	122.87
8	Aug	1.038	0.645	110.56
9	Sep	0.618	0.000	65.74
10	Oct	1.187	0.400	126.57
11	Nov	1.170	0.336	125.35
12	Dec	1.158	0.703	124.25

Detected anomalies

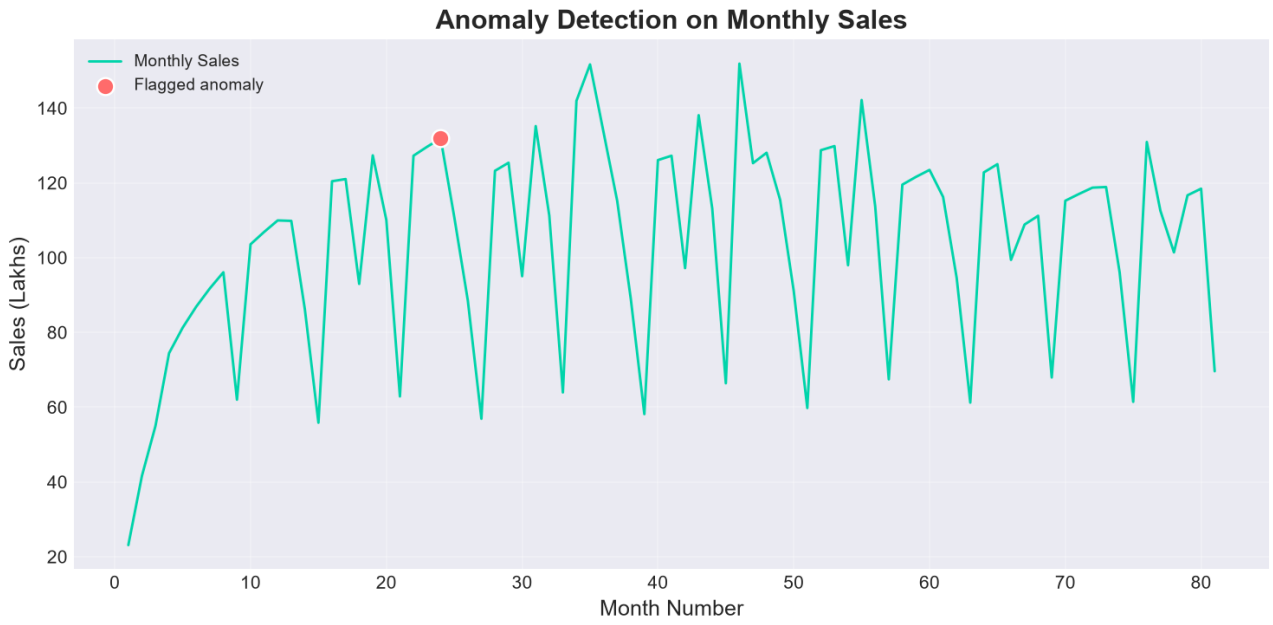
Month	Name	Sales	Expected	Score	Direction	Severity
24	Dec	131.87	104.01	3.021	spike	low



Monthly sales with a rolling average



Monthly gross margin with rolling average



Anomaly detection on monthly sales

3. Forecast Outlook

Model selection

Metric	Value
Selected model	Ensemble (Top 3)
Validation RMSE	8.02
Validation MAPE	5.95%
Forecast horizon (months)	6
Residual std. dev.	7.74
Methodology	Rolling-origin cross-validation (4 origins x 6-month horizon); final model(s) re-fitted on all 81 observations; prediction interval from a 2,000-draw residual bootstrap.

Point forecast with 95% prediction interval

Month	Point forecast	Lower 95%	Upper 95%
82	119.94	98.66	132.36
83	120.13	98.85	132.56
84	120.36	99.08	132.79
85	113.65	92.37	126.08
86	97.34	76.06	109.77
87	73.16	51.88	85.58

Ensemble composition

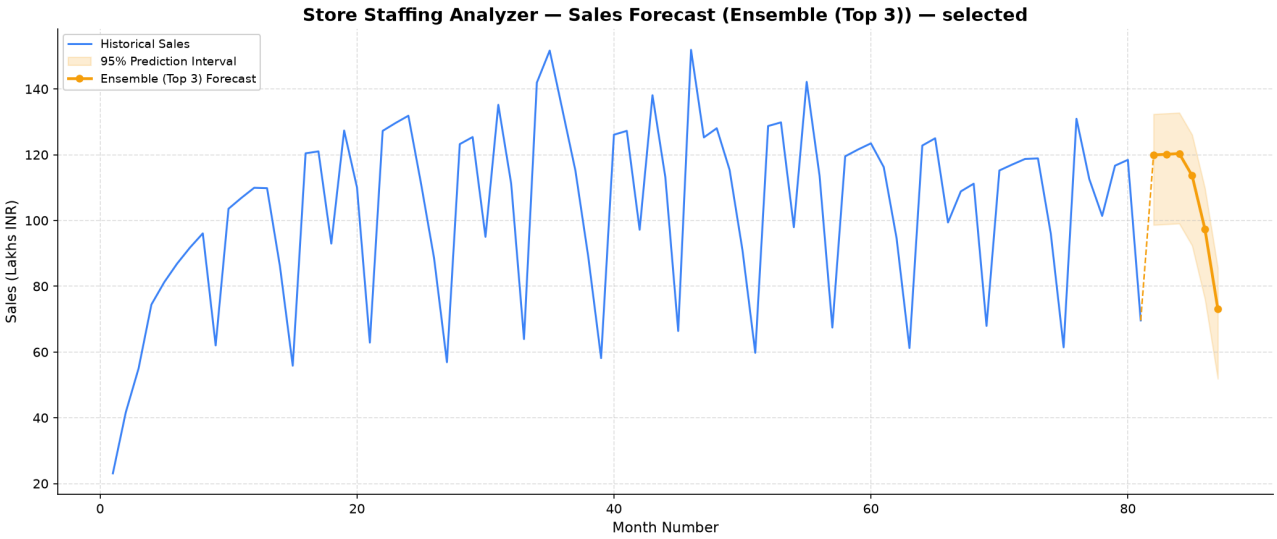
Model	Weight	RMSE	MAE	MAPE
Holt-Winters	42.7%	9.40	7.78	7.27%
Seasonal Naive	37.9%	10.58	6.34	5.61%
Holt (Damped Trend)	19.4%	20.70	15.54	18.55%

Model leaderboard

Model	RMSE	MAE	MAPE	MASE	R2	Rank
Holt-Winters	9.40	7.78	7.27%	0.30	0.789	1
Seasonal Naive	10.58	6.34	5.61%	0.24	0.732	2
Holt (Damped Trend)	20.70	15.54	18.55%	0.60	-0.024	3
Moving Average	20.82	16.08	18.80%	0.62	-0.036	4
Linear Regression	25.97	18.08	22.33%	0.69	-0.612	5
Naive	46.61	42.76	37.76%	1.64	-4.193	6
Random Forest	54.11	51.24	46.42%	1.96	-5.997	7

Walk-forward backtest

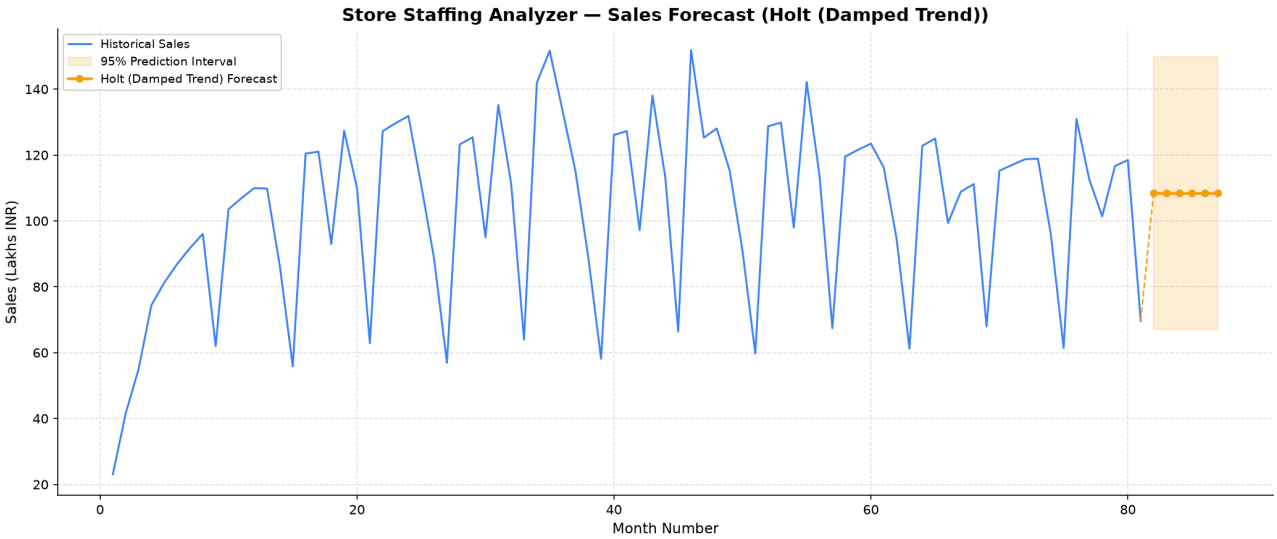
Origin month	Actual	Predicted	Error
58	61.21	73.14	-11.93
64	67.93	74.32	-6.38
70	61.43	69.02	-7.60
76	69.61	75.34	-5.73



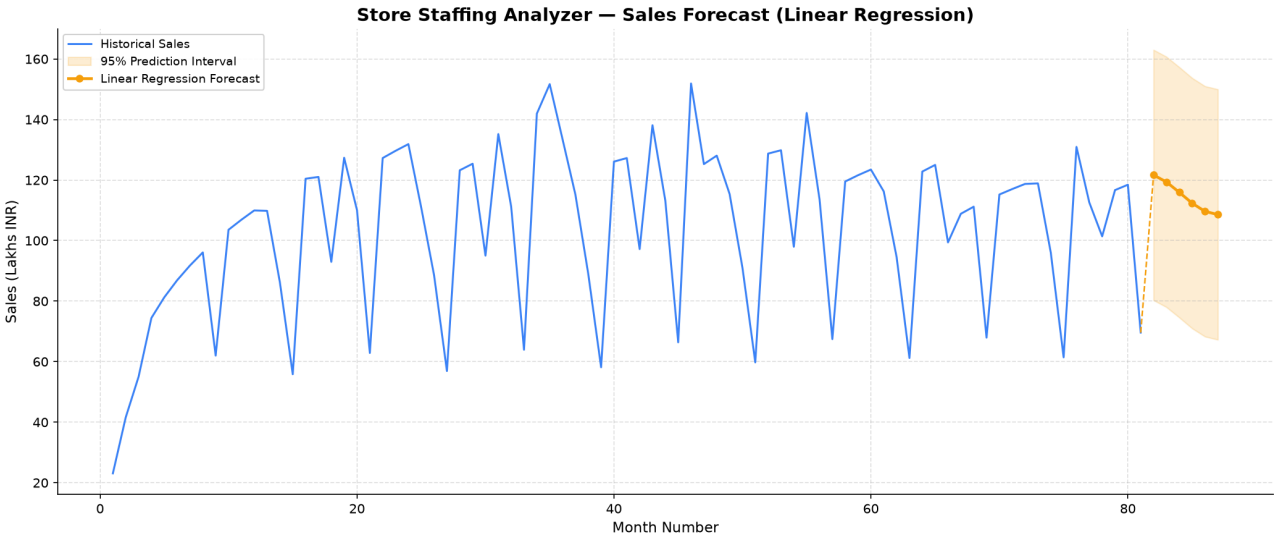
Forecast - forecast ensemble top 3



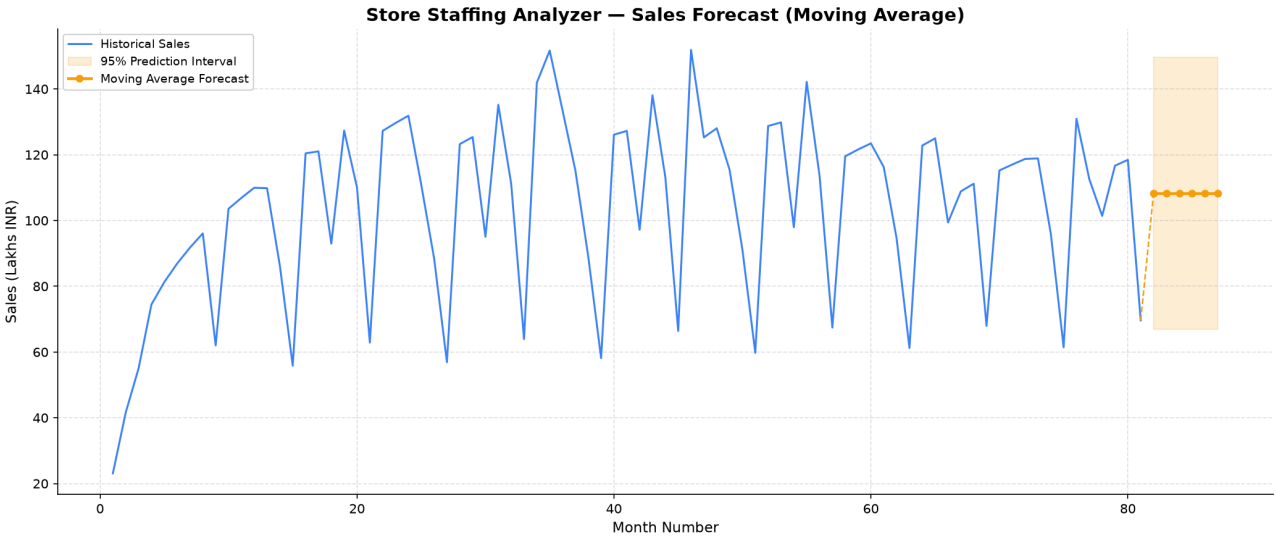
Forecast - forecast holt-winters



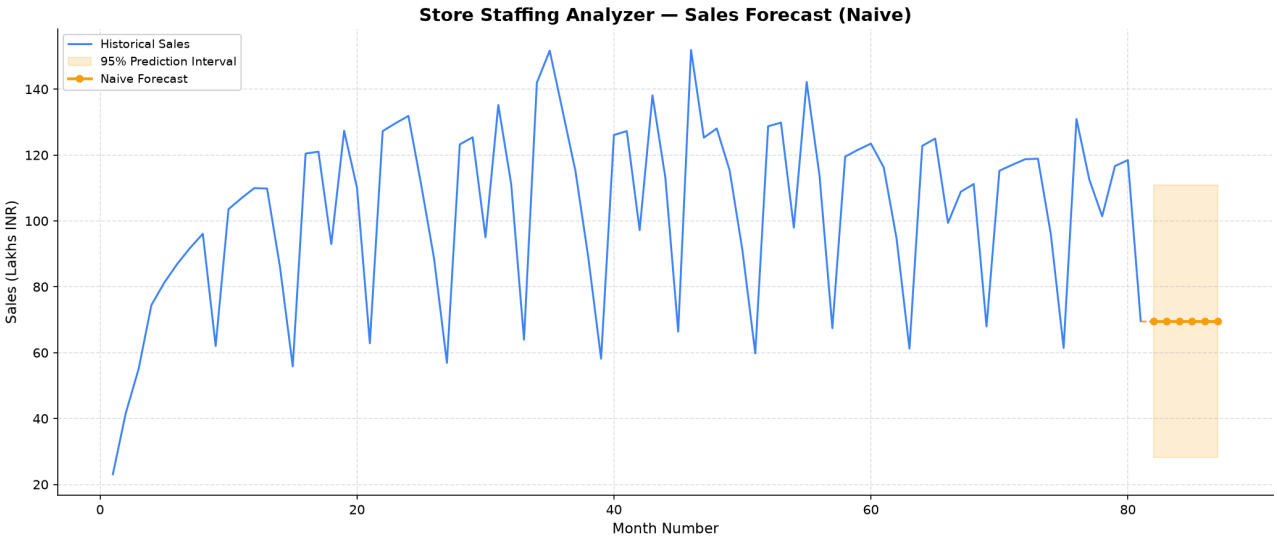
Forecast - forecast holt damped trend



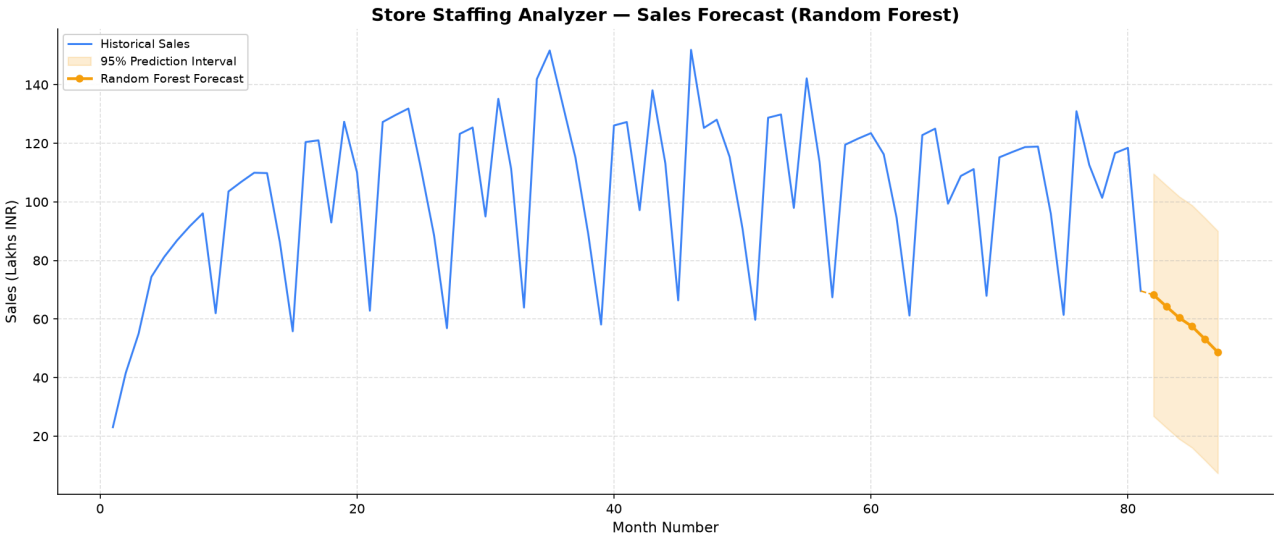
Forecast - forecast linear regression



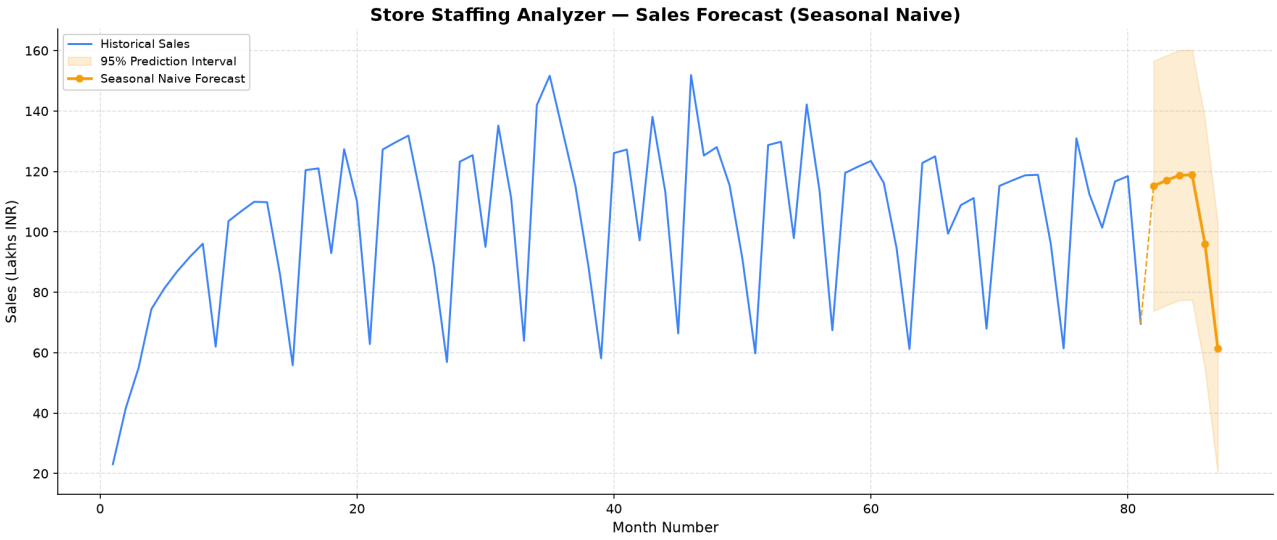
Forecast - forecast moving average



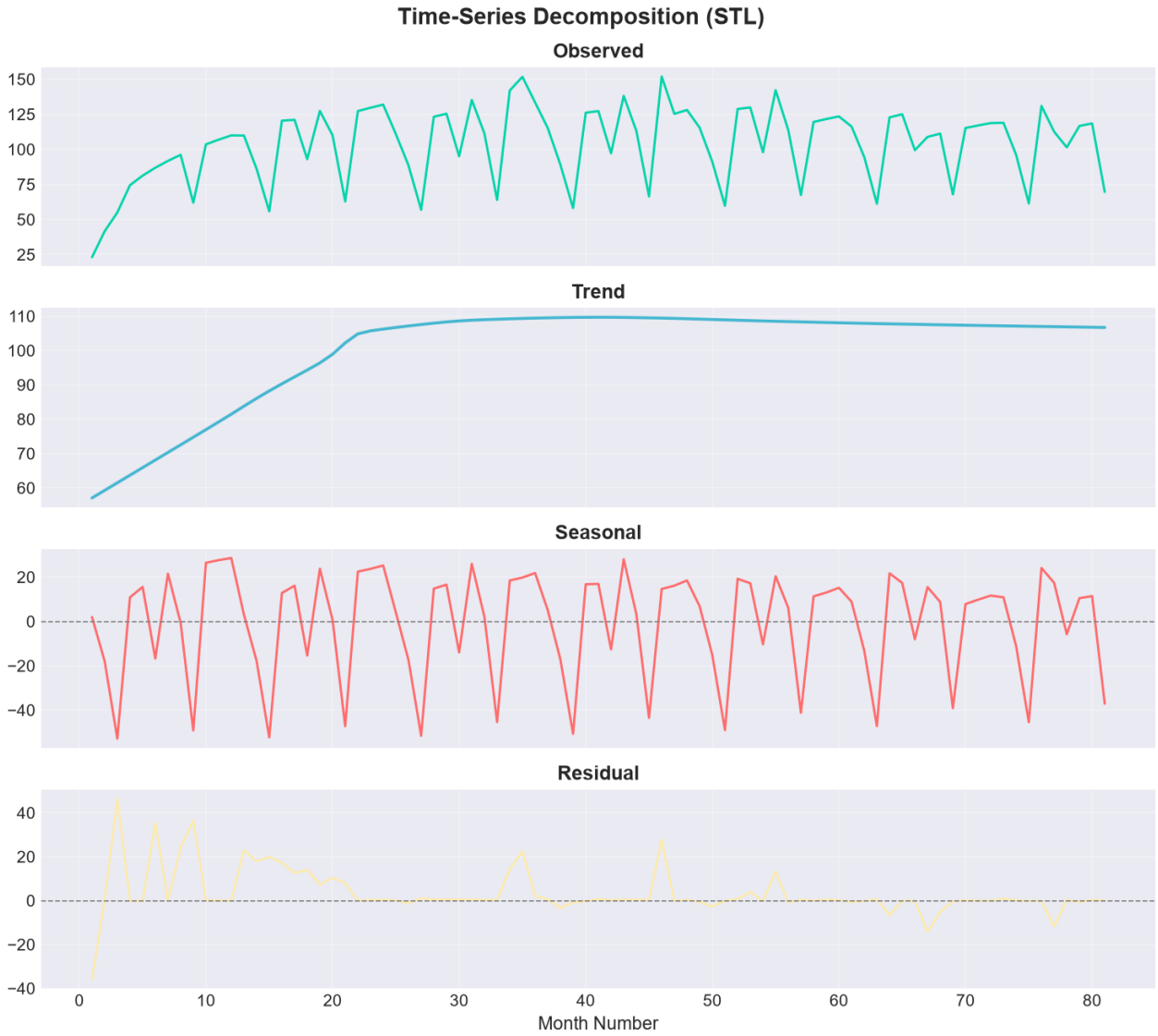
Forecast - forecast naive



Forecast - forecast random forest



Forecast - forecast seasonal naive



Time-series decomposition (trend / seasonal / residual)

4. Staffing Recommendation

Recommended action

Hire Immediately (8 from month 1)

CONFIDENCE: MEDIUM

Decision summary

Metric	Value
Recommended action	Hire Immediately (8 from month 1)
Recommended scenario	B
Confidence level	MEDIUM
Runner-up option	Scenario D: Conditional Hire (trigger >= 114.09 L sales)

Expected 6-month financial impact

Metric	Value
Total sales (6 months)	Rs. 702.52 L
Total gross margin (6 months)	Rs. 172.13 L
Total staff cost (6 months)	Rs. 179.13 L
Average monthly gross margin	Rs. 28.69 L
Margin advantage over runner-up	Rs. 0.00 L
Margin advantage vs do-nothing	Rs. 8.08 L

5. Scenario Comparison

Scenario A

Don't Hire (Stay at 7)

Scenario A - Don't Hire (Stay at 7)

Metric	Value
6-month gross margin	Rs. 164.06 L
Difference vs best scenario	Rs. -8.08 L
Total 6-month sales	Rs. 644.57 L
Total 6-month staff cost	Rs. 158.23 L
Average monthly margin	Rs. 27.34 L
P10 margin	Rs. 151.91 L
P50 margin	Rs. 164.06 L
P90 margin	Rs. 176.20 L
P(beats do-nothing baseline)	50.0%
P(margin > 0)	100.0%

Risk assessment

LOW financial risk but HIGH operational risk. Existing staff may burn out; customer service quality could decline during peak months.

Scenario B

Hire Immediately (8 from month 1)

Scenario B - Hire Immediately (8 from month 1)

Metric	Value
6-month gross margin	Rs. 172.13 L
Difference vs best scenario	Rs. 0.00 L
Total 6-month sales	Rs. 702.52 L
Total 6-month staff cost	Rs. 179.13 L
Average monthly margin	Rs. 28.69 L
P10 margin	Rs. 159.99 L
P50 margin	Rs. 172.13 L
P90 margin	Rs. 184.28 L
P(beats do-nothing baseline)	72.7%
P(margin > 0)	100.0%

Risk assessment

MODERATE risk. Higher fixed cost from day one, but the new hire ramps up productivity over 6 months. If sales dip, margins shrink.

Scenario C

Hire After 2 Months

Scenario C - Hire After 2 Months

Metric	Value
6-month gross margin	Rs. 168.26 L
Difference vs best scenario	Rs. -3.87 L

Metric	Value
Total 6-month sales	Rs. 680.58 L
Total 6-month staff cost	Rs. 172.03 L
Average monthly margin	Rs. 28.04 L
P10 margin	Rs. 154.93 L
P50 margin	Rs. 168.26 L
P90 margin	Rs. 181.59 L
P(beats do-nothing baseline)	61.7%
P(margin > 0)	100.0%

Risk assessment

LOW-MODERATE risk. Saves 2 months of extra salary while still bringing a new hire on board before peak season pressure builds.

Scenario D

Conditional Hire (trigger >= 114.09 L sales)

Scenario D - Conditional Hire (trigger >= 114.09 L sales)

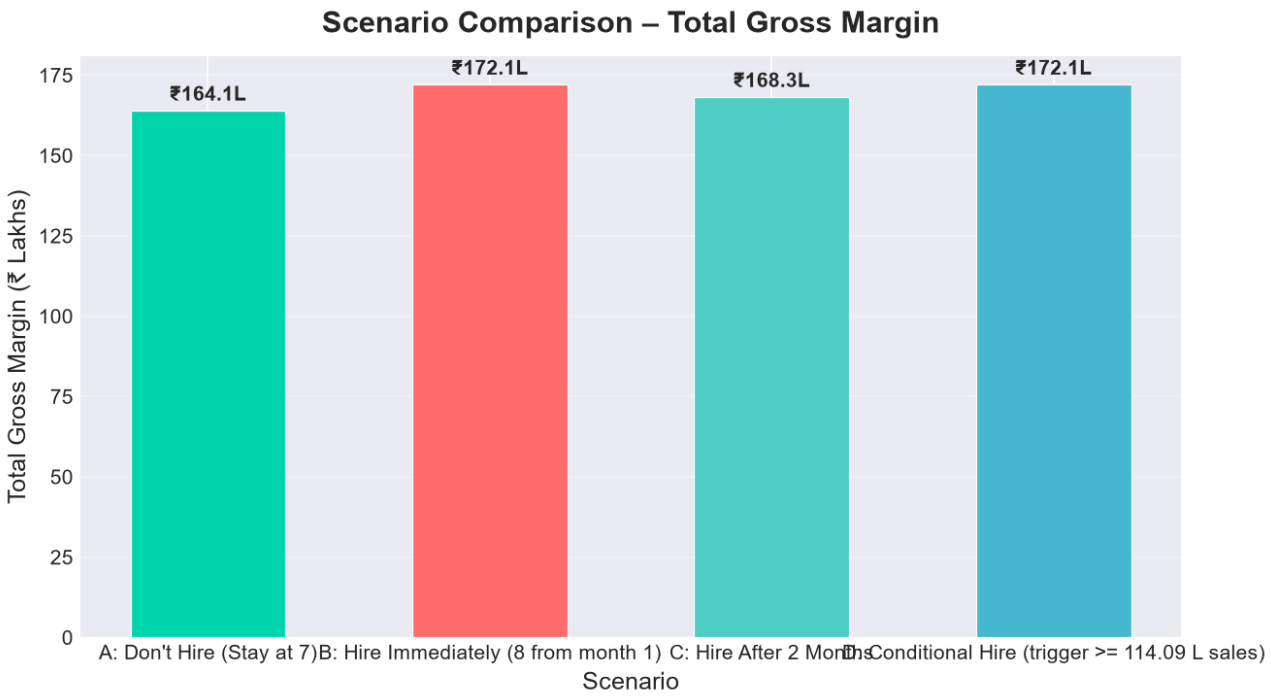
Metric	Value
6-month gross margin	Rs. 172.13 L
Difference vs best scenario	Rs. 0.00 L
Total 6-month sales	Rs. 702.52 L
Total 6-month staff cost	Rs. 179.13 L
Average monthly margin	Rs. 28.69 L
P10 margin	Rs. 159.99 L
P50 margin	Rs. 172.13 L
P90 margin	Rs. 184.28 L
P(beats do-nothing baseline)	72.7%
P(margin > 0)	100.0%

Risk assessment

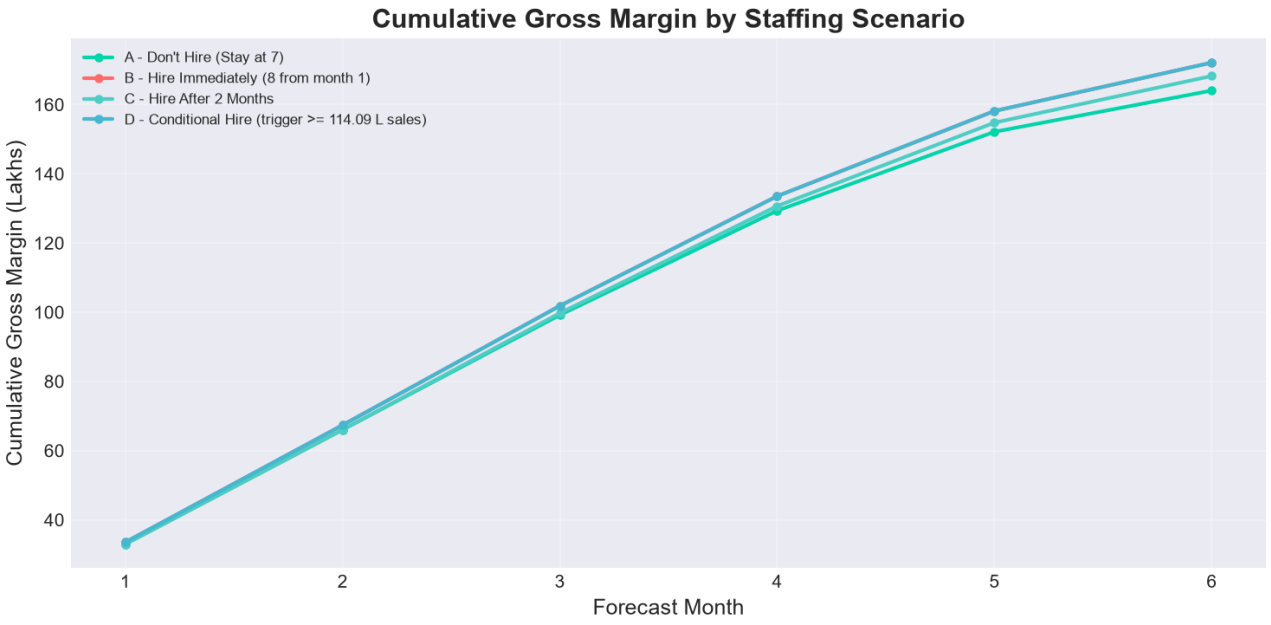
LOW risk. Data-driven decision minimises cost if sales are weak, but may result in delayed hiring when the market picks up.

Recommended scenario B - month-by-month detail

Month	Employees	Sales	Gross margin	Staff cost
1	8	127.81	33.52	30.39
2	8	128.57	33.86	30.43
3	8	129.93	34.47	30.50
4	8	123.78	31.70	30.19
5	8	108.03	24.61	29.40
6	8	84.41	13.98	28.22



Total 6-month gross margin by scenario



Cumulative gross margin path by scenario

6. Productivity & Diminishing Returns

Productivity summary

Diminishing returns detected: Yes. Capacity-optimum headcount: 9.

Reading the headcount optimum

The optimiser searches the headcount range the store has actually operated (plus one hire) and treats the marginal contribution as constant, so a persistently positive marginal pushes the answer to the top of that range. The scenario comparison answers the practical question -- hire one, wait, or hold -- under an explicit ramp-up curve.

Performance by headcount

Headcount	Avg sales	Avg sales / employee	Avg gross margin	Months observed
3	23.10	7.70	1.39	1
5	41.59	8.32	3.72	1
7	104.59	14.94	26.06	70
8	110.00	13.75	25.50	9

Marginal contribution of each additional hire

From heads	To heads	Marginal sales / added	Change in sales / employee	Diminishing
3	5	9.25	0.62	No
5	7	31.50	6.62	No
7	8	5.42	-1.19	Yes

Expected margin by target headcount

Headcount	6-month margin	Avg monthly margin	Margin / employee
3	114.53	19.09	6.36
4	126.91	21.15	5.29
5	139.29	23.22	4.64
6	151.68	25.28	4.21
7	164.06	27.34	3.91
8	172.13	28.69	3.59
9	178.69	29.78	3.31

Controlled marginal productivity (OLS with trend and season controls)

Metric	Value
Marginal sales per employee	Rs. 11.25 L
Marginal margin contribution	Rs. 2.06 L
p-value	0.0001
Model R2	0.8901
Observations	80
Model	first_difference

First-difference OLS: month-on-month changes in sales are regressed on month-on-month changes in headcount with a trend and month-of-year controls, so the secular growth of the store is differenced out before the staffing effect is measured.

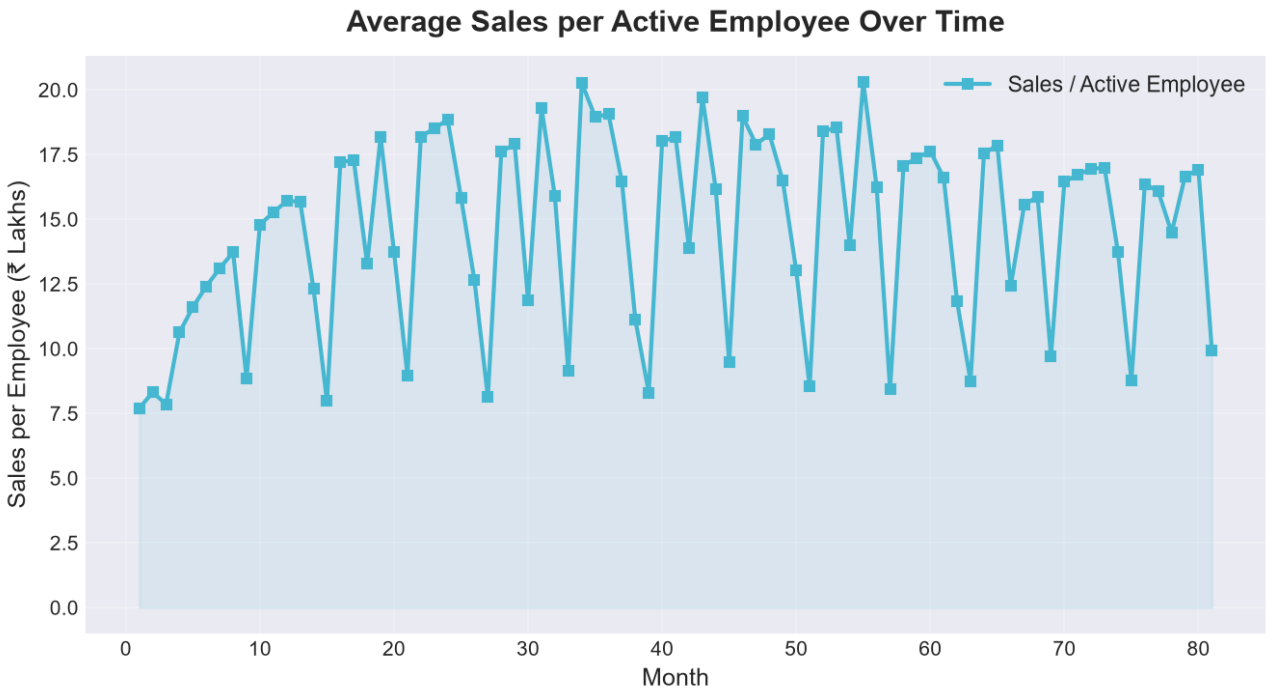
Workforce roster

Salesperson	Total sales	Avg monthly sales	Months active	Tenure (months)	Still active
sale sp7	1,166.54	21.21	55	55	No

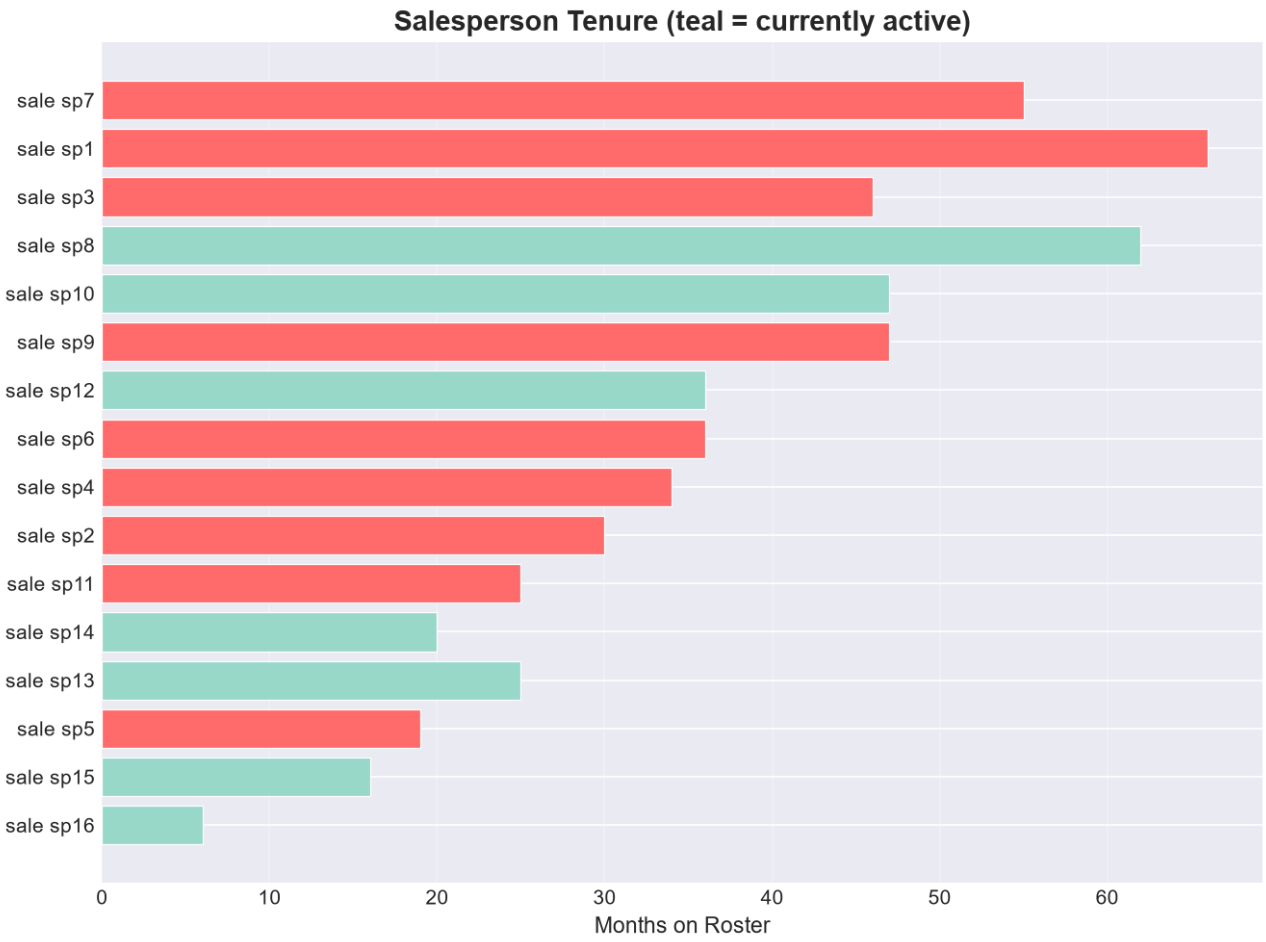
Salesperson	Total sales	Avg monthly sales	Months active	Tenure (months)	Still active
sale sp1	1,087.87	16.48	66	66	No
sale sp3	899.74	19.56	46	46	No
sale sp8	798.50	12.88	62	62	Yes
sale sp10	772.50	16.44	47	47	Yes
sale sp9	647.75	13.78	47	47	No
sale sp12	557.08	15.47	36	36	Yes
sale sp6	500.11	13.89	36	36	No
sale sp4	487.72	14.34	34	34	No
sale sp2	351.18	11.71	30	30	No
sale sp11	282.95	11.32	25	25	No
sale sp14	262.31	13.12	20	20	Yes
sale sp13	216.39	8.66	25	25	Yes
sale sp5	183.29	9.65	19	19	No
sale sp15	115.07	7.19	16	16	Yes
sale sp16	46.79	7.80	6	6	Yes

Attrition statistics

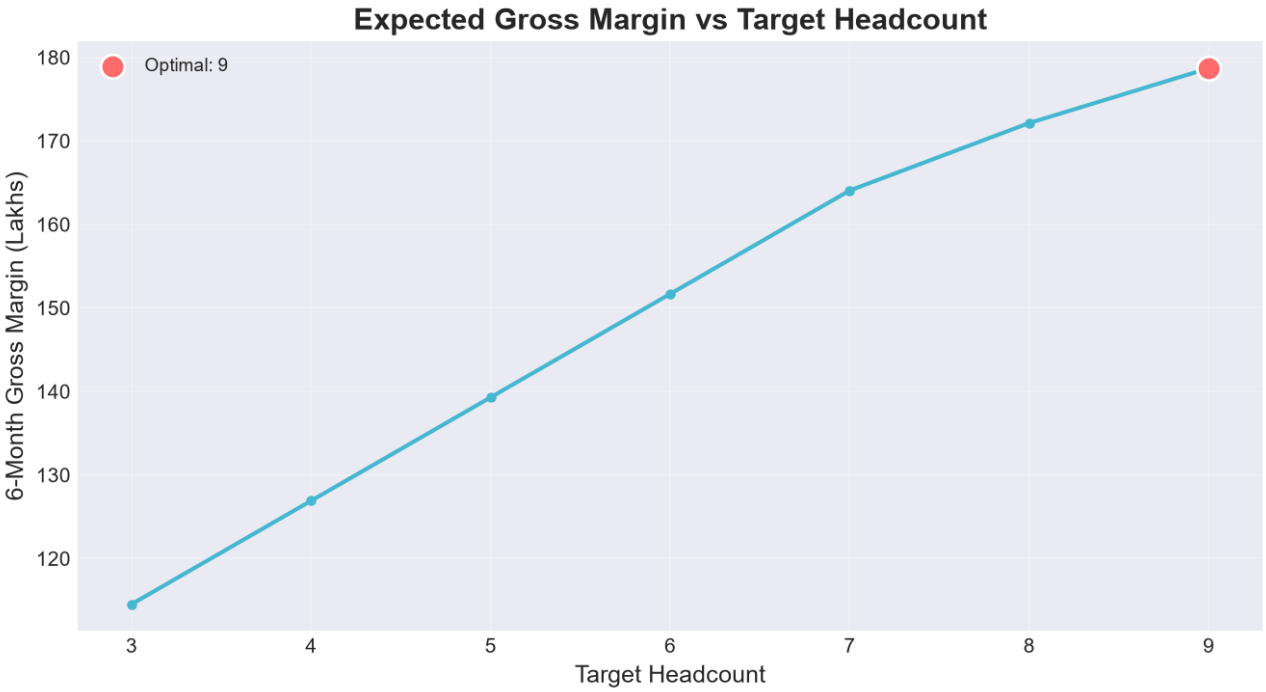
Metric	Value
Total leavers	9
Total hires	16
Average tenure of leavers (months)	39.8
Leavers in last 12 months	1
Retention rate	100.0%



Average sales per active employee



Salesperson tenure



Expected margin vs target headcount

7. Risk & Sensitivity

Tornado analysis - one-at-a-time sensitivity

Variable	Low	High	Low margin	High margin	Swing
Sales volume	x0.8	x1.2	Rs. 16.23 L	Rs. 34.84 L	Rs. 18.61 L
Monthly salary	2	4.5	Rs. 32.53 L	Rs. 15.03 L	Rs. 17.50 L
COGS rate	0.42	0.58	Rs. 33.80 L	Rs. 17.26 L	Rs. 16.54 L
Commission rate	0.02	0.09	Rs. 28.63 L	Rs. 21.40 L	Rs. 7.24 L

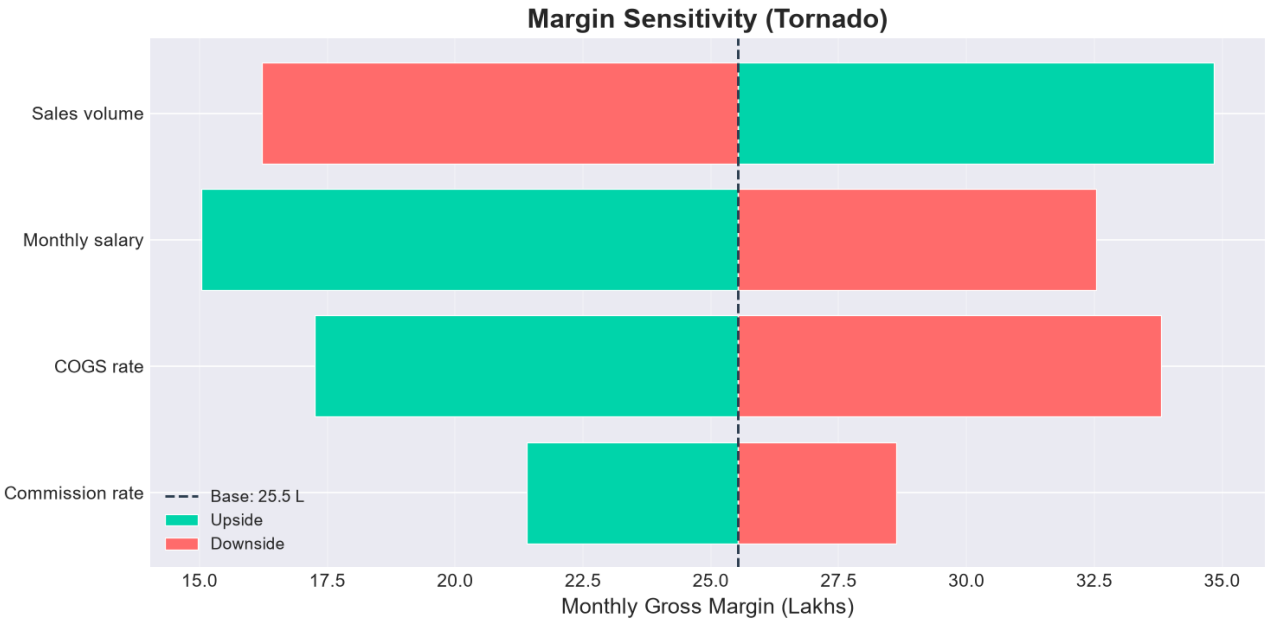
Monte-Carlo margin distribution

Metric	Value
P10 margin	Rs. 130.12 L
P50 margin (median)	Rs. 146.13 L
P90 margin	Rs. 162.37 L
Mean margin	Rs. 146.27 L
Standard deviation	Rs. 12.59 L
P(margin > 0)	100.0%
Simulations	5,000

Decision stability

Metric	Value
Leading scenario	Don't Hire (Stay at 7)
Margin over runner-up	Rs. 12.00 L
Decisive	Yes

The winning scenario is separated from the runner-up by more than half a standard deviation, so the decision is robust to the simulated demand and cost uncertainty.



Margin sensitivity (tornado)

8. Payback & ROI

Investment case

Metric	Value
Months to payback	1.00
Return on investment	44.9%
Investment per month	Rs. 3.00 L
Net benefit	Rs. 8.08 L
Total contribution over horizon	Rs. 8.08 L
Ever pays back	Yes

Cumulative contribution of the recommended hire

Month	Monthly contribution	Cumulative contribution
1	0.54	0.54
2	0.80	1.34
3	1.30	2.65
4	1.56	4.20
5	1.81	6.01
6	2.06	8.08

9. Business Rule

Recommended business rule

Adaptive Staffing Rule: Maintain a 3-month rolling average of sales per employee. If the rolling average exceeds Rs.16.48 L, hire one additional employee. If it falls between Rs.12.79 L and Rs.16.48 L, hold current headcount and replace departures. If it drops below Rs.12.79 L, do NOT replace departing staff. Optimal target headcount based on historical margins: 7 employees.

Rolling-average trigger

Metric	Value
Window (months)	3
Hire above	16.48
Hold band	12.79 to 16.48
Do not replace below	12.79
Current rolling average	14.51

Threshold-based decision rules

Condition	Action
sales_per_emp >= 17.60	HIRE an additional employee.
11.88 <= sales_per_emp < 17.60	HOLD current headcount.
sales_per_emp < 11.88	DO NOT replace departing employee.

Why this beats 'always replace'

- The 'always replace' rule ignores current sales levels. During low-sales months the extra salary erodes margin.
- The adaptive rule ties staffing decisions to actual revenue performance, avoiding unnecessary costs in lean months and scaling up when the market supports it.
- Run `backtest_staffing_rules` for the measured backtested saving rather than an assumed percentage.

Backtest interpretation

Replaying both policies over 81 months, the adaptive rule produced Rs.3.00 L of gross margin (0.1%) more than always replacing, and avoided overstaffing in 17 month(s) (21 % of the period).

Backtest - adaptive rule vs always replace

Policy	Total margin	Avg monthly margin
Always replace	Rs. 2,068.11 L	Rs. 25.53 L
Adaptive rule	Rs. 2,071.11 L	Rs. 25.57 L
Improvement	Rs. 3.00 L	0.1%

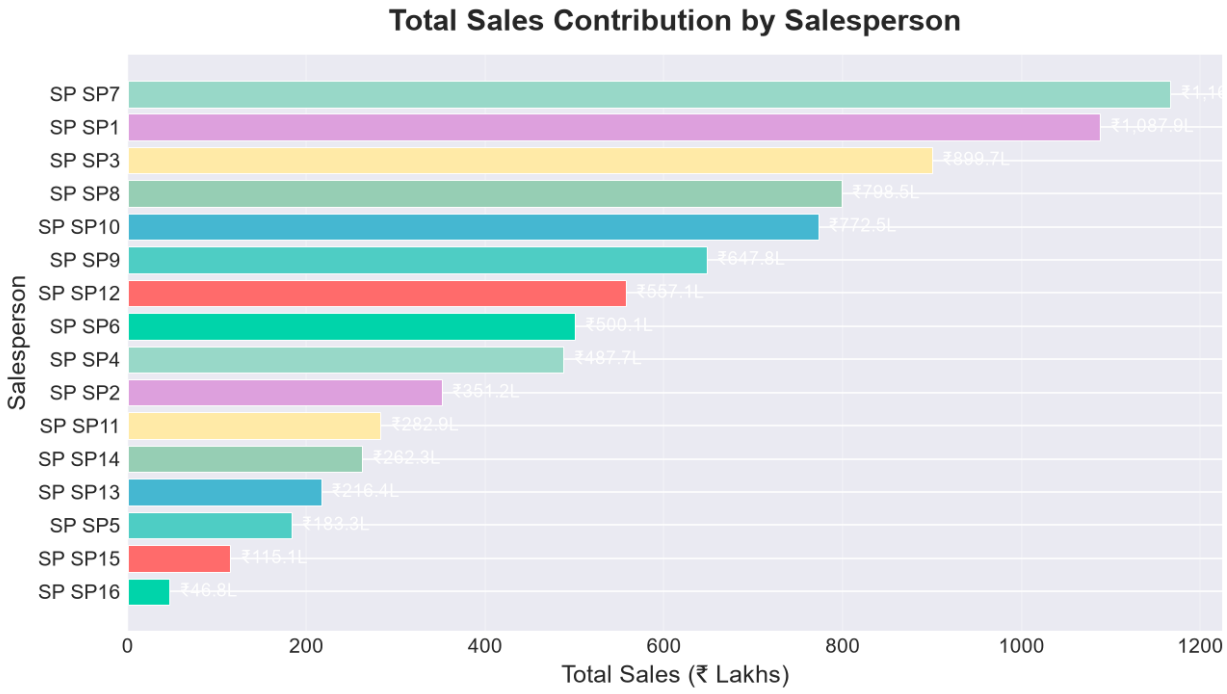
Backtest coverage

Metric	Value
Months compared	81
Months of overstaffing avoided	17
Share of overstaffing avoided	21.0%

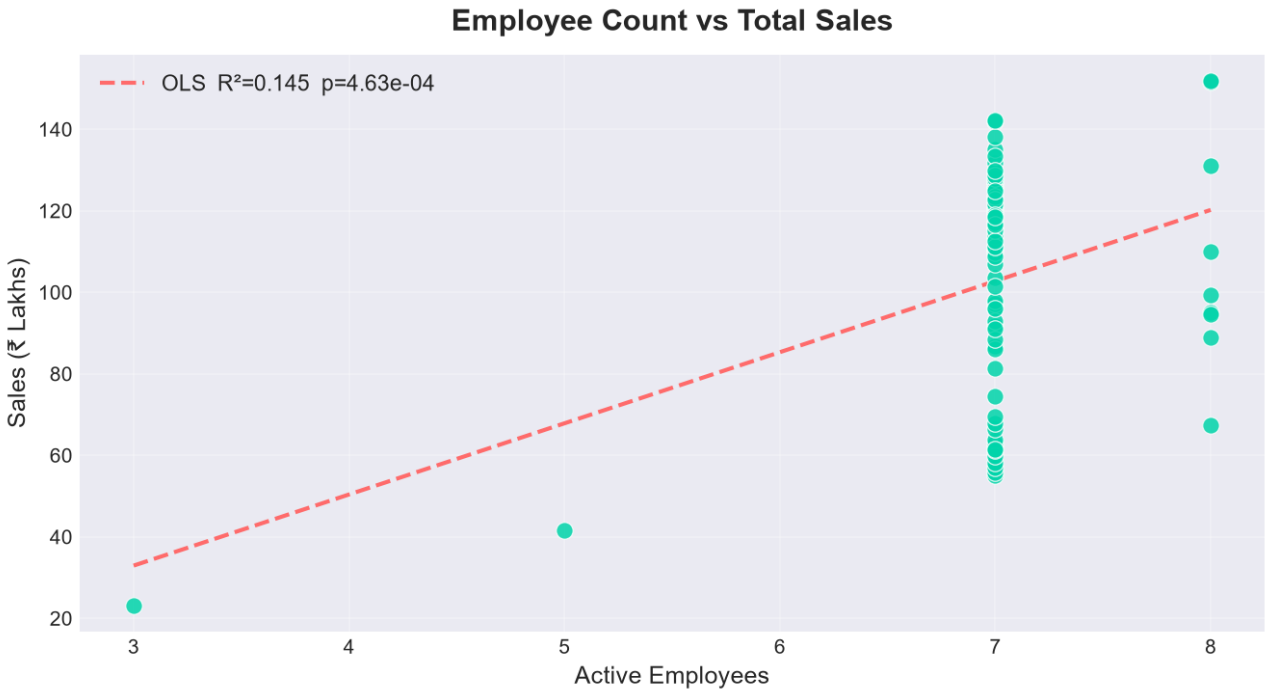
10. Appendix

Business assumptions

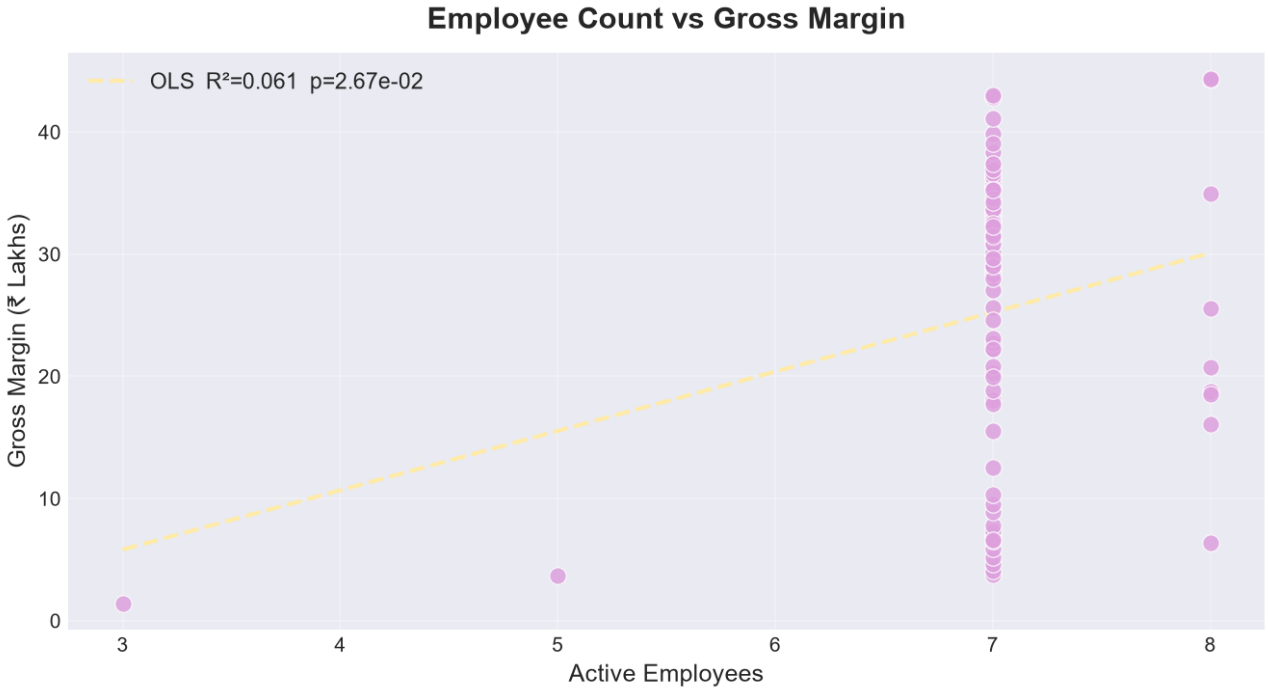
- Customer traffic and footfall remain stable.
- No significant hiring or recruitment costs are incurred.
- No training-period productivity loss beyond the modelled ramp-up (70% -> 100% over 6 months).
- Monthly salary stays fixed at Rs. 3.00 Lakhs per employee.
- Commission remains fixed at 5.0% of total sales.
- Cost of goods sold stays at 50% of sales.
- Economic conditions and consumer demand remain similar.
- No major competitor entry or exit during the projection period.
- Store operating hours remain unchanged.
- The pricing and assortment strategy remains consistent.
- Sales are treated as exogenous: the analysis prices staffing decisions against realised demand.



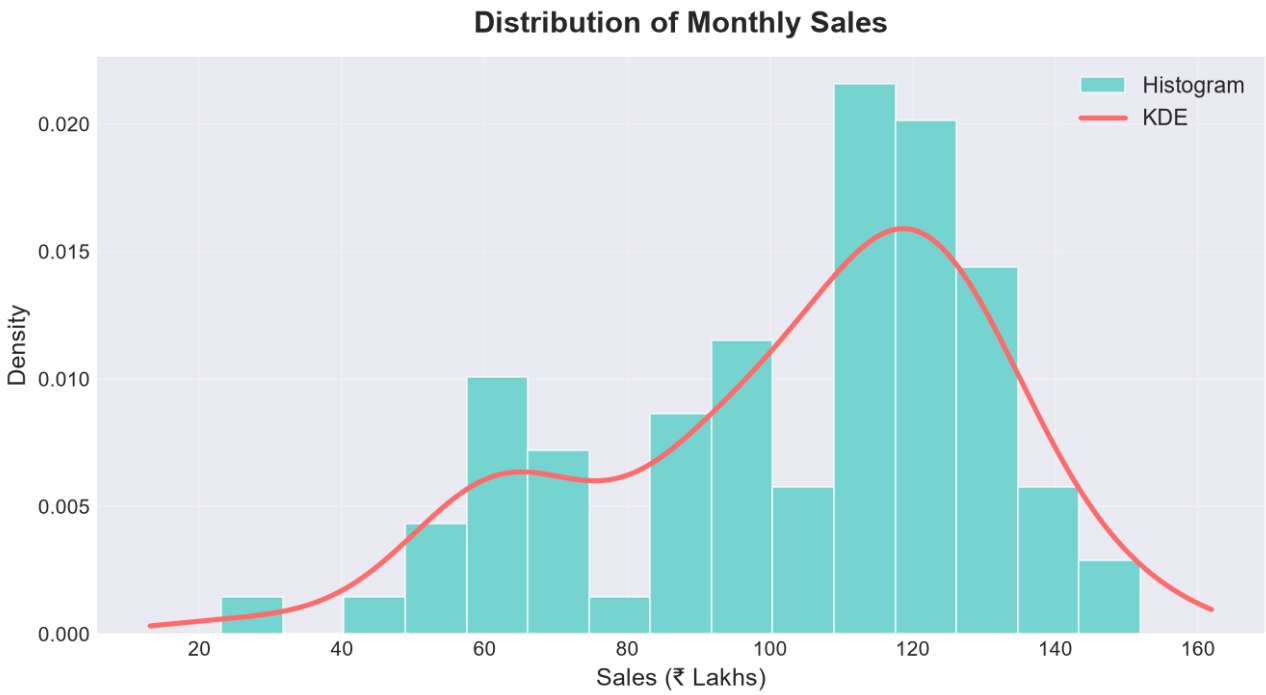
Salesperson productivity



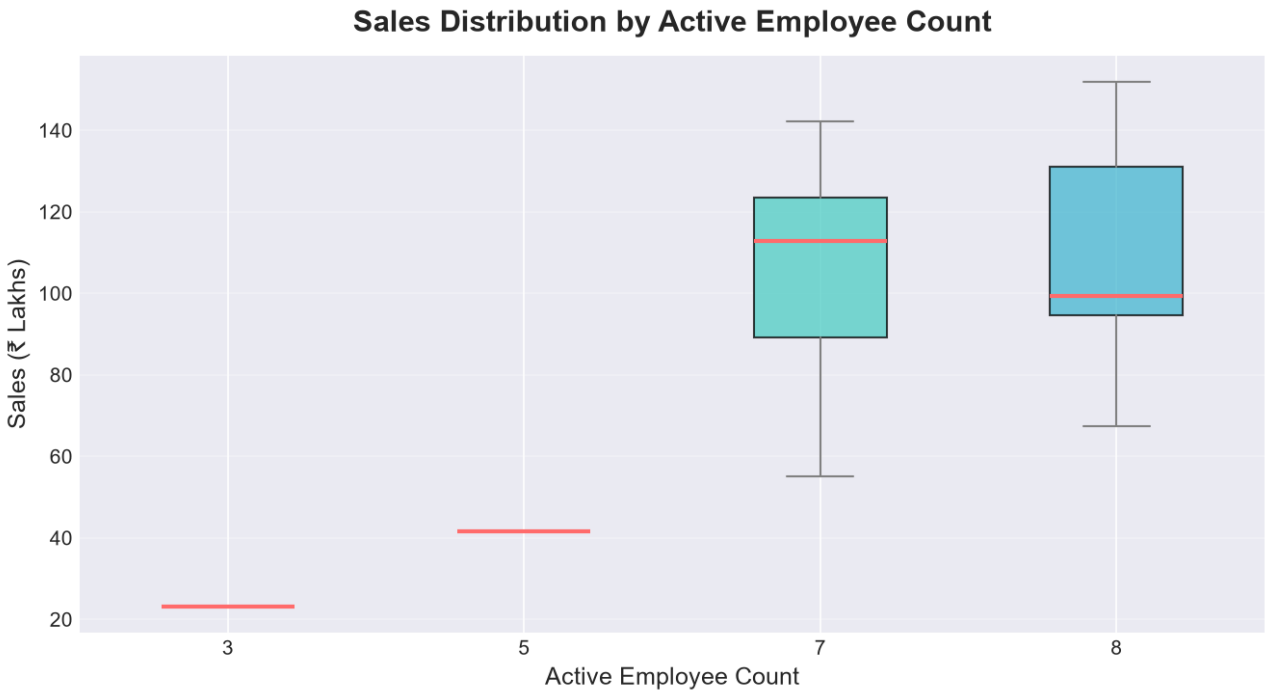
Employee count vs sales



Employee count vs margin

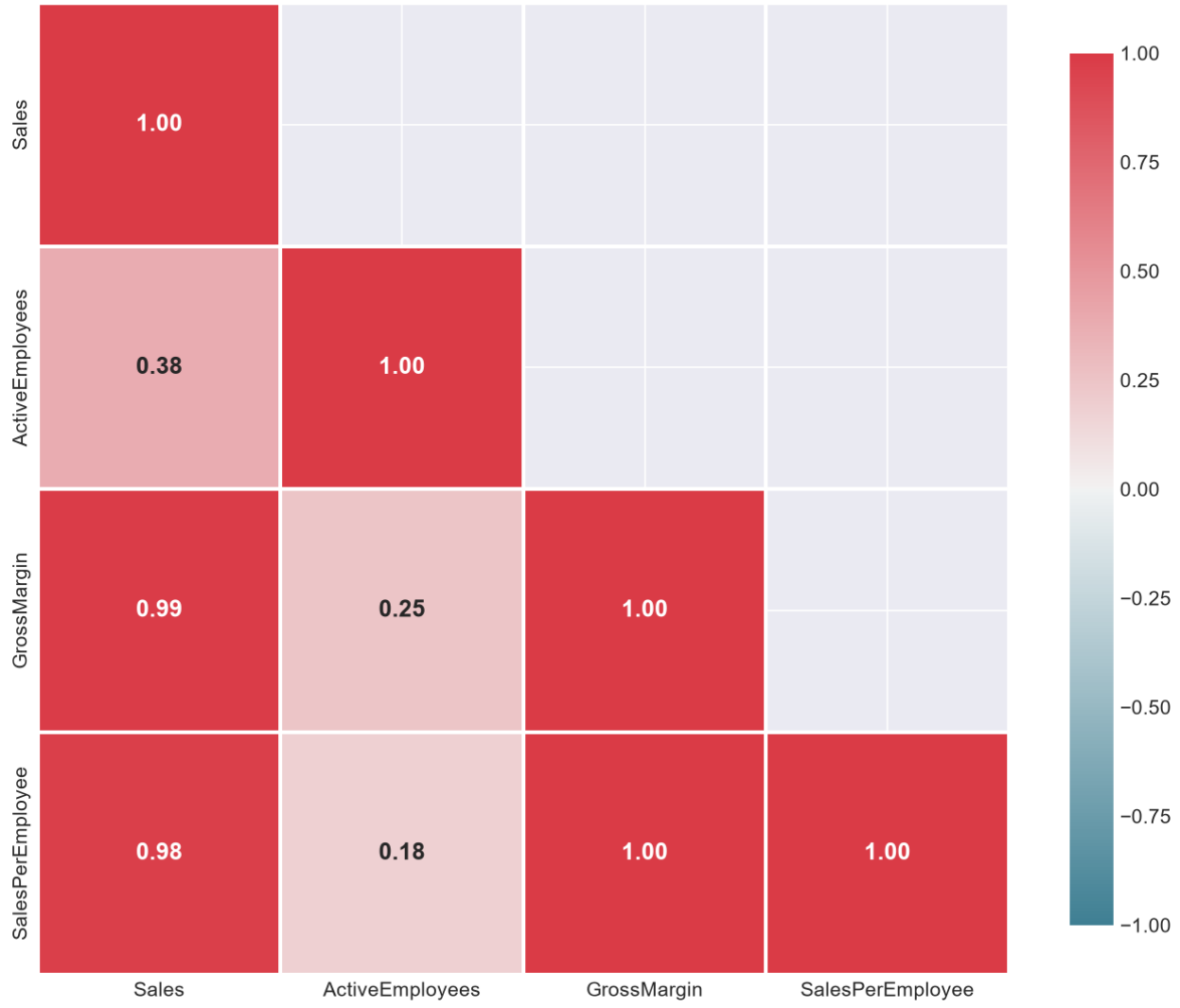


Sales histogram

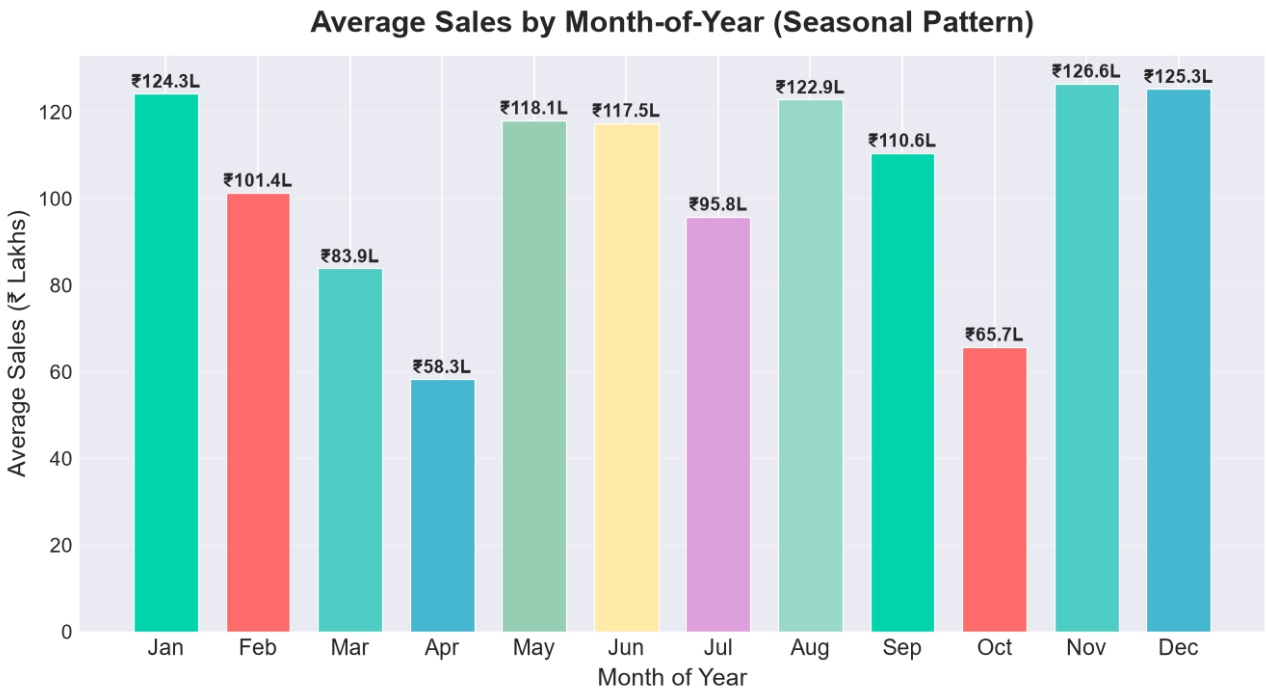


Sales boxplot

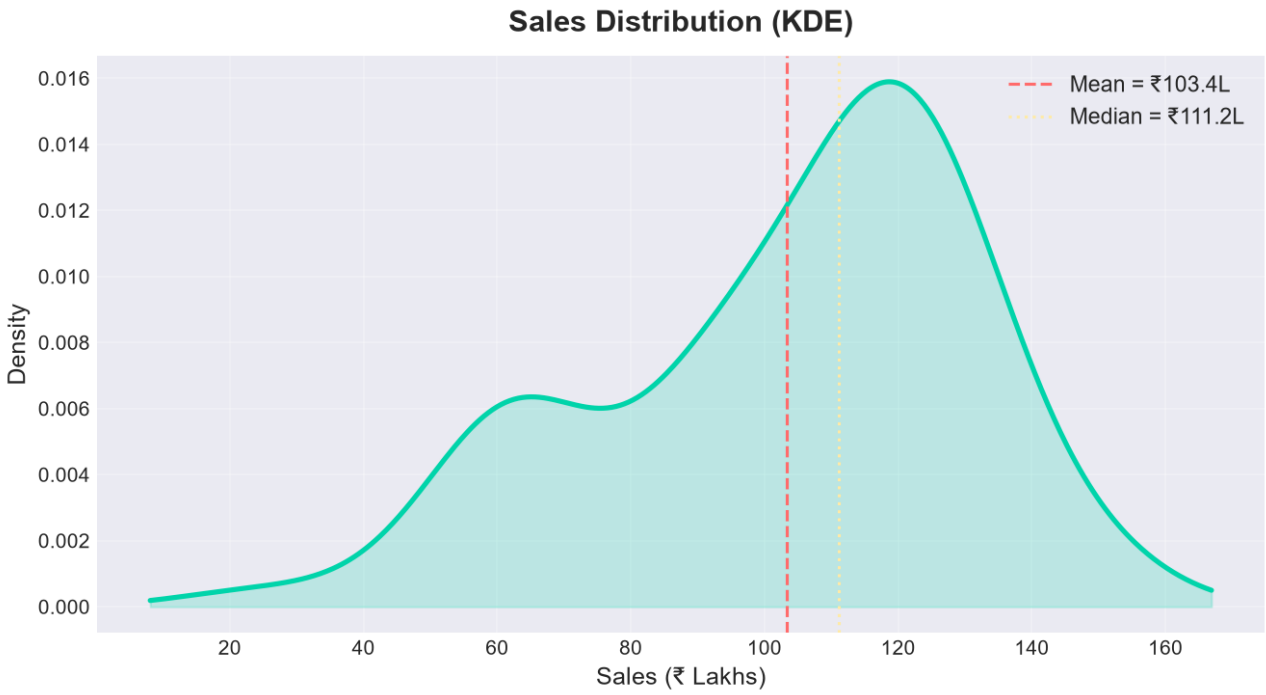
Correlation Matrix – Key Staffing Metrics



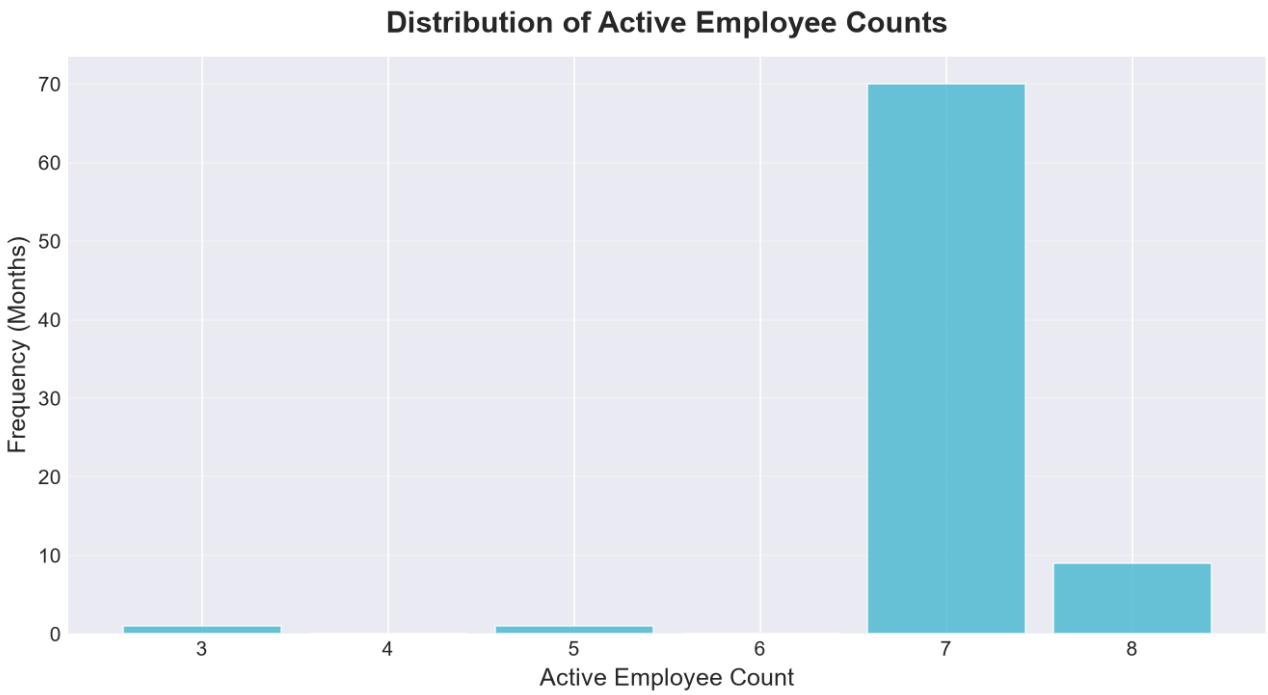
Correlation heatmap



Monthly seasonality



Sales distribution



Employee count distribution



Rolling metrics